

ALLEN JOHN DARIN J

Aspiring Aerospace Engineer

CONTACT

Phone:

+91 8438518756

Email:

jajdarin@gmail.com

Linkedin:

www.linkedin.com/in/allen-jd

Address:

Namakkal, India

EDUCATION

B.E. in Aerospace Engineering

Mahendra Engineering College,
India

2023 - 2027

**Higher Secondary Certificate
(Maths-Biology Stream)**

C.S.I Matriculation Higher
Secondary School , India
2022 - 2023

SKILLS

- Agile Team Leadership
- Project Management
- CAD / CAE Tools
- Full Stack Web Developer
- Control system design
- Aerospace Structural Analysis

SUMMARY

Dedicated Aerospace Engineering student with interdisciplinary expertise spanning propulsion systems, structural analysis, and AI-enabled aerospace applications. Proven ability to develop simulation-driven engineering solutions and contribute to research-oriented projects. Actively engaged in space technology initiatives, conferences, and innovation challenges.

EXPERIENCE

INTERN

KUN AEROSPACE. – Chennai, India | Dec 2025 – Jan 2026

- Worked across Management Review (MR) and Quality departments
- Assisted in documentation review, process tracking, and audit-related activities within MR
- Gained hands-on exposure to aerospace quality inspection procedures, standards, and compliance practices
- Developed practical understanding of quality systems and continuous improvement processes

INTERN / TRAINEE

ISRO Propulsion Complex (IPRC) – Tirunelveli, India | Jan 2025

- Gained insights into liquid propulsion systems and cryogenic engines, enhancing understanding of aero engine concepts.
- Observed assembly, integration, and testing processes of propulsion systems for PSLV and GSLV.
- Learned propellant storage and combustion techniques, applying problem-solving skills in rocket propulsion.
- Acquired knowledge of mission planning, quality control, and safety protocols in aerospace engineering.

CERTIFICATIONS

- NASA – Open Science Training (Essentials & 101)
- CATIA V5 (Beginner to Advanced – Udemy)
- SOLIDWORKS Sustainability Associate - Dassault Systems
- MATLAB Essential Training - LinkedIn
- Aha! Product Management Professional Certificate - LinkedIn
- Enhancing soft skills and personality - NPTEL
- Space Flight Mechanics - NPTEL

PROFESSIONAL MEMBERSHIPS

- Student Member, ISTE – Indian Society for Technical Education
 - Student Member, Royal Aeronautical Society (RAeS)
 - Student Member, Institution of Mechanical Engineers (IMechE)
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PROJECTS

Eclipse-Resilient Heat Battery

- Designed a passive/semi-passive spacecraft thermal management system using Phase Change Materials (PCM) to mitigate extreme orbital temperature cycling.
- Developed a heat battery concept enabling latent heat storage during solar exposure and controlled heat release during eclipse phases. Focused on improving spacecraft energy efficiency, reliability, and mission endurance.

AI-Based Space Debris Collision Prediction and Avoidance

- Developed an AI-powered framework to predict potential collisions between active satellites and orbital debris using TLE (Two-Line Element) data.
- Implemented orbital trajectory simulations with Python (Skyfield, Pandas, Matplotlib) and built a 3D visualization tool.

AI-Based Urban Air Traffic Management System for UAV

- Built an AI-powered system to manage UAV airspace by dynamically assigning altitudes using fuzzy logic and multi-agent coordination. Integrated real-time UAV tracking on a web map using Flask and Leaflet.
- Developed backend modules and REST APIs for UAV state management, altitude assignment, and real-time data updates.

Smart Adaptive Hybrid Joints

- Designed an innovative hybrid joint mechanism capable of adapting to variable mechanical loads and vibrations.
 - Integrated smart elements to dynamically respond to stress, temperature, and fatigue conditions.
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LANGUAGES

- English
- Tamil

ACHIEVEMENTS

- Selected for Smart India Hackathon (SIH 2025 & 2024) Grand Finale – Team Leader & Core Member
 - Participated in NASA Space Apps Challenge
 - Selected for State-Level Youth Astronomy & Space Science Congress (YASSC 2025) – Team Leader & Core Member
 - Finalist – Inter-College Hackathon (2025 & 2026)
 - Participant, IDE Bootcamp 2026 (Phase 1) – Organized by AICTE and Ministry of Education's Innovation Cell.
 - Participated in Bharatiya Antariksh Hackathon conducted by ISRO
 - Participant in World Space Week 2025 Webinars - Day & Night Space Foundation (DANSF)
 - Received Multiple Awards and Prizes in Technical Paper Presentations.
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Research / Publications

- Selected for AstroSat International Conference 2026 as a Speaker
UR Rao Satellite Centre (URSC) – ISRO
Paper: Next-Generation Cryogenic Cooling Concepts for High-Energy Detectors in Space Missions
 - Presented Oral Paper at International Conference on Aerospace and Aeronautical Engineering - Inovine Meetings
Paper: AI-Powered Predictive Maintenance Framework for Aircraft Engines using Flight Data Recorder Logs
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Patents

- Prototype Development of Smart Adaptive Hybrid Joints for Aerospace Vehicles and System Thereof
Indian Patent Application No. 202541012674
Status: Published (2025)
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